Sense, Show, Act: Placement-Aware Sensing, Visualization, and Interaction on Interactive Surfaces

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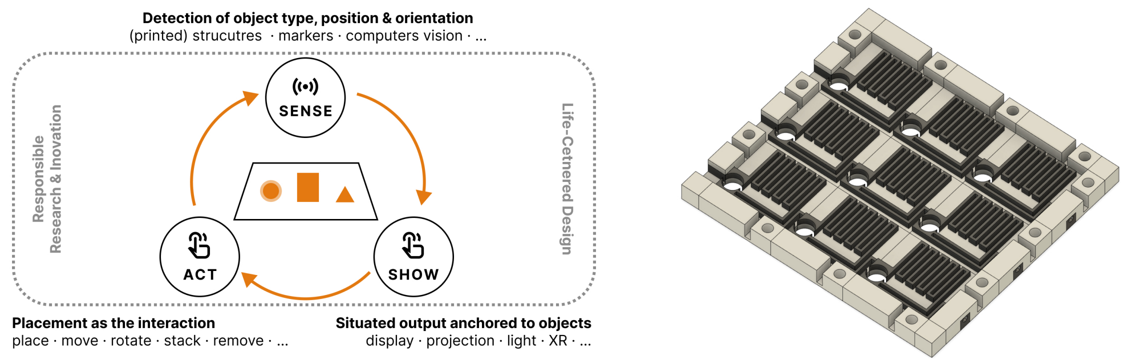

(a) (b)

Fig. 1. **(a)** The three coupled strands of the workshop form a continuous loop around placement-aware interactive surfaces: *sensing* detects object type, position, and/or orientation; *visualization* anchors situated output to physical objects; and *interaction* treats placement itself as input. A cross-cutting sustainability and life-centered design lens (dashed) frames all three; **(b)** Render of a prototype that enables new forms of unobtrusive object detection through simple 3D-printed circuits. This acts as one example of novel techniques that we aim to discuss in this workshop.

Interactive surfaces and spaces increasingly need to know *what* objects are placed on or near them and *where* and *how* they sit – and to turn that knowledge into meaningful interaction. This challenge spans three tightly coupled strands: *placement-aware sensing* of object type, position, and orientation via hardware-, marker-, or vision-based approaches; *placement-aware visualization* through displays, light, projection, or XR, anchored to physical objects; and *placement-aware interaction*, in which placing, moving, altering,

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and removing objects are deliberate activities. Choices in any one strand constrain the other two. Each carries material, energetic, and social aspects that are still underexplored. This half-day workshop brings together researchers and practitioners across all three strands to jointly map the design space of placement-aware interactive surfaces and to explore novel techniques. Through position papers, hands-on demos, and structured ideation combining insight harvesting/combination with design studio techniques under lifecycle constraints, participants will develop application concepts, identify open challenges, and initiate collaborations toward a shared research agenda, published proceedings, novel methods, and cross-community follow-up projects.

CCS Concepts: • **Human-centered computing → Interaction design**.

Additional Key Words and Phrases: interactive surfaces, tangible interaction, object recognition, placement detection, situated visualization, extended reality, digital fabrication, sustainable HCI

ACM Reference Format:
Lukas A. Flohr, Julia Dominiak, Julian Rasch, Bastian Pfleging, Paweł W. Woźniak, and Sebastian S. Feger. 2026. Sense, Show, Act: Placement-Aware Sensing, Visualization, and Interaction on Interactive Surfaces. In *Proceedings of ACM Interactive Surfaces and Spaces (ISS '26).* ACM, New York, NY, USA, 8 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Background and Rationale

Recognizing objects and their precise placement on and around interactive surfaces has been a recurring ambition of the ISS community since its tabletop origins [10, 12]. Recent years have brought rapid but also fragmented progress across three strands (sense, show, act; Figure 1 (a)) that, we argue, must be designed together.

1.1 Three Coupled Research Strands

The sensing approach determines which placement semantics can be detected and at what fidelity. That fidelity constrains which visualizations are feasible and which interactions are reliable. The intended interaction vocabulary sets requirements back onto sensing. Yet the strands are advanced largely by separate sub-communities, and systems are frequently designed from one strand outward, inheriting avoidable limitations in the other two.

1.1.1 Placement-Aware Sensing. Sensing determines object type, position, and/or orientation. Fabrication-embedded approaches – such as capacitive and conductive markers [19, 20, 29], RFID/NFC tagging [15], and passive impedance-based structures [6] – now compete and combine with marker- and vision-based recognition using cameras above, below, or within the surface [5, 7, 13], whose current challenges and opportunities are themselves under active reflection [17]. Emerging sensing techniques further enable new applications and interactions. The prototype shown in Figure 1(b), for instance, uses 3D-printed conductive filament to form a simple circuit: placing an object with one or more conductive pins changes the circuit's overall resistance, enabling unobtrusive object sensing. Such prototypes exemplify the kind of innovation we want to discuss at this workshop.

1.1.2 Placement-Aware Visualization. Visualization responds with situated output anchored to physical objects and their spatial context: on-display rendering around tangibles [24], top-down and through-surface projection [11, 16], and head-mounted, see-through, or handheld XR – from AR overlays anchored to physical referents [3, 8, 14, 21] through VR settings in which tracked physical objects serve as tangible proxies [18, 22].

1.1.3 Placement-Aware Interaction. Interaction treats placement itself as an expressive vocabulary: placing, nudging, rotating, stacking, or removing an object is not incidental to interaction but *is* the interaction [1, 2, 28].

1.2 A Cross-Cutting Sustainability Lens

A further dimension cuts across all three strands: the material, energetic, and social cost of placement awareness [4]. Battery-powered smart tangibles create charging overhead and e-waste, embedded electronics render printed objects effectively unrecyclable, always-on cameras consume energy continuously and implicate bystanders who never opted in. Recent work on unmaking, decomposable devices, and recyclable electronics [30] shows that interactive systems need not carry these costs – but the community lacks a shared vocabulary for weighing such trade-offs in placement-aware systems. Prior workshops have addressed sustainable fabrication broadly [23] but not connected to an integrated view on placement sensing, visualization, and interaction. We therefore adopt a sustainability and life-centered design lens [9, 25, 27] as a recurring evaluative dimension applied to discussed techniques and concepts.

1.3 Goals and Relevance to ISS

The workshop pursues four goals: (1) provide the foundation for discussing and mapping an integrated design space connecting placement-aware sensing, visualization, and interaction, with lifecycle and more-than-human considerations as dimensions; (2) identify open research challenges at the seams between the strands; (3) generate specific cross-disciplinary application concepts through structured ideation; and (4) initiate lasting collaborations, follow-up publications, and joint project proposals.

The topic sits at the historical core of the conference (tangibles on interactive tabletops) while engaging its current frontiers (fabrication, XR, spatial computing) and its explicitly invited themes of new methods, sustainable innovation, and ethical considerations in interactive surfaces and spaces.

2 Pre-Workshop Plans

Ahead of the workshop we will establish its online presence, recruit participants across the relevant communities, ensure accessibility, and distribute materials on the timeline below.

2.1 Website and Communication

Upon acceptance, we will launch a workshop website with the call, submission instructions, dates, contacts, program, and – after selection – accepted papers and pre-workshop materials. A dedicated e-mail address and mailing list will support communication. We will provide title, description, and URL to the ISS Web Chairs and coordinate promotion with the Publicity Chairs.

2.2 Recruiting

We will distribute the call through the organizers' networks across the ISS, CHI, TEI, UIST, AVI, IUI, ISMAR, VRST, and MuC communities; relevant mailing lists (e.g., CHI-announcements), social media, and direct invitations to groups active in fabrication-embedded sensing, vision-based recognition, situated visualization, XR, tangible interaction, and sustainable HCI. The complementary background of the organizing team (see Section 6) gives us direct access to the target communities.

2.3 Accessibility and Inclusion

The call and website will follow SIGCHI accessibility guidance; distributed PDFs will be tagged and screen-reader accessible, and pre-workshop videos captioned. We will invite participants to communicate accessibility needs at submission and upon acceptance, and coordinate requirements (e.g., transcription, reserved seating) with the Accessibility Chairs. Accepted papers and materials will be distributed at least two weeks before the workshop. For community members unable to attend in person, the call, accepted papers, video previews (where authors consent), and results will be available asynchronously via the website.

2.4 Timeline for Participants

Table 1. Proposed timeline (2026), aligned with the ISS recommendation to notify authors at least five days before the early-bird registration deadline (Oct. 23).

Date	Milestone
Aug. 10	Website and call for papers live
Sep. 15	Short Paper and demo submission deadline
Sep. 30	Notification to authors
Oct. 30	Camera-ready due; distribution of materials begins
Nov. 9	Pre-workshop reading package sent (2 weeks ahead)
Nov. 23	Workshop day (day before the main conference)

3 Workshop Format (In-Person, Half-Day)

The workshop is a half-day, in-person event for 15–20 participants (minimum 10), initially featuring short paper presentations and hands-on demonstrations that feed a structured ideation process. Interaction is the organizing principle: presentations are deliberately short, and every activity generates material for the second part: a hands-on design studio.

3.1 Submission Categories

We invite (a) position papers and (b) short technical or work-in-progress papers, both 2–4 pages excluding references in the single-column ACM format, plus optional (c) demo abstracts of 1–2 pages, encouraging participants to bring prototypes, printed objects, sensing rigs, or visualization demos. Submissions should address at least one focus area – placement-aware sensing, visualization, or interaction, including XR scenarios in which physical placement shapes augmented, mixed, or virtual experiences – and are encouraged to reflect on lifecycle, sustainability, or more-than-human implications.

3.2 Structured Ideation

We combine two established methods: Insight Harvesting and Design Studio.

3.2.1 Insight Harvesting. First, all participants capture insights on structured cards (e.g., observation; focus area(s) touched; material, energetic, or social cost). This keeps the audience active during presentations and yields a great collection of insights that are clustered on a shared wall before the design studio.

3.2.2 Design Studio. After clustering, mixed teams – each seeded with at least one participant per focus area – work on application briefs (e.g., kitchen, makerspace, museum, in-vehicle, control room, board gaming), each carrying a how-might-we question with an explicit lifecycle constraint (e.g., "objects must be detected even when they are outside the field of view or obscured," "objects must remain sensible after five years of use," "no reliable power to objects"). Following a design studio approach [26], team members first sketch individually to develop initial ideas and expand the solution space. Teams then critique in a pin-up session along three rubric lenses – feasibility across the sensing–visualization–interaction pipeline, interaction quality, and lifecycle/externalities – before converging on one concept per team, grounded in harvested insights from different focus areas. Sketching individually before discussing as a group counters dominance effects and lowers the barrier for non-native English speakers.

3.3 Schedule

Table 2 shows the proposed half-day schedule. Organizers act as facilitators and timekeepers, one per breakout team.

Table 2. Proposed half-day schedule. The schedule may be adjusted based on the (number of) submissions and participants; it scales from 10 participants (2 teams) to 20 (4 teams).

Timing	Activity	Engagement
0:00–0:15	Welcome, goals, lightning introductions	Plenary, individual
0:15–0:45	Keynote bridging the three strands	Talk + Q&A
0:45–1:30	Paper/demo madness (3–5 min per paper/demo, by focus area)	Short talks, active listening
1:30–1:40	Short break	
1:40–2:10	Provocation: sustainability & life-centered design	Impulse talk
2:10–2:35	Insight wall clustering, affinity mapping	Plenary, standing
2:35–3:20	Ideation: Design Studio	Mixed teams of 4–6
3:20–3:45	Report-backs and plenary synthesis: design space & research agenda	Plenary discussion
3:45–4:00	Roadmap: publications, collaboration matchmaking, next steps	Plenary

3.4 Technical Capacity and Room Requirements

One room with movable tables (island seating for 3–4 teams), a projector with HDMI, power strips for demo tables, and 4–6 m of free wall or pin-board space. The organizers bring all ideation materials and a camera for documentation. Wi-Fi for participants; no special network requirements. Accessibility requirements will be confirmed with participants in advance and coordinated with the Accessibility Chairs.

4 Diversity and Language

The organizing team spans five institutions across three countries and complementary sub-communities (fabrication/sensing, visualization/XR, interaction design/prototyping, sustainable HCI). In participant selection, given comparable quality, we aim to balance career stage, institution and geography, gender, and focus-area coverage, and explicitly welcome practitioners alongside academics. Our methods are chosen for inclusion: lightning-length talks prevent seniority from dominating airtime; written insight harvesting and individual sketching give voice to participants less comfortable speaking in plenary or in English; and mixed-team rules ensure every sub-community is represented in every team. The life-centered design lens further broadens whose interests are considered – including non-users, bystanders, and environmental stakeholders.

5 Expected Outcomes

The workshop is designed to produce tangible results across three timelines:

Immediate: Accepted papers are published as open-access proceedings (see Section 7).

Short-Term: A workshop report synthesizes the design space of placement-aware sensing, visualization, and interaction – with lifecycle considerations as explicit dimensions – alongside a prioritized research agenda. In addition, the report documents the insight wall, clusters, and team sketches that will be photographed and digitized.

Medium-Term: We close with explicit matchmaking for follow-up publications and joint proposals (e.g., Horizon Europe), and will gauge interest in a follow-up edition or a journal special issue.

6 Organizers

Lukas A. Flohr is a professor of interactive media production at TH Rosenheim, with a PhD from Saarland University on context-based prototyping of human-machine interfaces. His research bridges UX, XR, human factors, and design thinking focusing on prototyping and evaluation methods for immersive and emerging interaction paradigms – offering a methods-driven, interdisciplinary perspective for the workshop's focus on interactive surfaces.

Julia Dominiak is an assistant professor at Lodz University of Technology, Poland, and currently a NAWA Bekker Programme fellow at TU Wien, Austria. Her research focuses on placement-based tangible interaction, spanning a capacitance-based toolkit for tangible prototyping and data physicalization, hybrid board games instrumented with EEPROM-tagged pieces, and reconfigurable workstations controlled through tangible interfaces. She runs the 3D printing and prototyping laboratory at Lodz University of Technology, contributing a fabrication and embedded sensing perspective to the workshop's exploration of placement-aware sensing and interaction.

Julian Rasch is a researcher and PhD Student at the Media Informatics Group at LMU Munich. His work focuses on XR, immersive interaction, and multi-user collaboration, including how virtual content can be situated in and connected to physical environments. He contributes an XR perspective to the workshop's exploration of placement-aware visualization and interaction across interactive surfaces and spaces.

Bastian Pfleging is professor for Ubiquitous Computing and Smart Systems at TU Bergakademie Freiberg, Germany. His research lies at the intersection of human-computer interaction and ubiquitous computing, with a focus on multimodal, context-aware, and intelligent interactive systems. His work addresses application domains including future mobility, human-AI and human-robot interaction, immersive technologies, and usable security.

Paweł W. Woźniak is professor for human-computer interaction and head of a research unit at TU Wien. His research focuses on the intersection of technologies, sport, and wellbeing, exploring everyday experiences of physical activity and personal informatics to design technologies that support wellbeing.

Sebastian S. Feger is a Professor at TH Rosenheim and director of the study program Smart Interactive Media. His research focuses on Human-Computer Interaction (HCI), physical computing, and the design of interactive systems at the intersection of hardware, software, and the physical world. He explores modern fabrication methods, including digital fabrication, rapid prototyping, and novel approaches for creating interactive and immersive experiences.

7 Publishing Plans

Subject to authors' consent, accepted workshop papers are planned to be published as an open-access volume in the CEUR Workshop Proceedings series or on arXiv, ensuring citability and long-term availability. The camera-ready deadline will be set so the volume is available before the workshop day. The workshop report and digitized ideation outcomes will additionally be published on the website.

References

[1] Till Ballendat, Nicolai Marquardt, and Saul Greenberg. 2010. Proxemic Interaction: Designing for a Proximity and Orientation-Aware Environment. In *ACM International Conference on Interactive Tabletops and Surfaces (ITS '10)*. ACM, 121–130. doi:10.1145/1936652.1936676

[2] Patrick Baudisch, Torsten Becker, and Frederik Rudeck. 2010. Lumino: Tangible Blocks for Tabletop Computers Based on Glass Fiber Bundles. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '10)*. ACM, 1165–1174. doi:10.1145/1753326.1753500

[3] Melanie Berger, Aditya Dandekar, Regina Bernhaupt, and Bastian Pfleging. 2021. An AR-Enabled Interactive Car Door to Extend In-Car Infotainment Systems for Rear Seat Passengers. In *Proceedings of the 2021 Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '21)*. doi:10.1145/3411763.3451589

[4] Eli Blevis. 2007. Sustainable Interaction Design: Invention & Disposal, Renewal & Reuse. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '07)*. ACM, 503–512. doi:10.1145/1240624.1240705

[5] Mustafa Doga Dogan, Ahmad Taka, Michael Lu, Yunyi Zhu, Akshat Kumar, Aakar Gupta, and Stefanie Mueller. 2022. InfraredTags: Embedding Invisible AR Markers and Barcodes Using Low-Cost, Infrared-Based 3D Printing and Imaging Tools. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM. doi:10.1145/3491102.3501951

[6] Bhaskar Dutt, Yujie Chen, Daniel Ashbrook, and Valkyrie Savage. 2025. Impedius: A Signal-Space Multiplexing Technique Using Individual Elements of Impedance for Chained Passive Sensors. In *Proceedings of the Nineteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '25)*. ACM, 1–11. doi:10.1145/3689050.3704949

[7] Sebastian S. Feger, Lars Semmler, Albrecht Schmidt, and Thomas Kosch. 2022. ElectronicsAR: Design and Evaluation of a Mobile and Tangible High-Fidelity Augmented Electronics Toolkit. *Proc. ACM Hum.-Comput. Interact.* 6, ISS, Article 587 (Nov. 2022), 22 pages. doi:10.1145/3567740

[8] Lukas A. Flohr, Joseph S. Valiyaveettil, Antonio Krüger, and Dieter Wallach. 2023. Prototyping Autonomous Vehicle Windshields with AR and Real-Time Object Detection Visualization: An On-Road Wizard-of-Oz Study. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23)* (Pittsburgh, PA, USA). ACM, 2123–2137. doi:10.1145/3563657.3596051

[9] Elisa Giaccardi and Johan Redström. 2020. Technology and More-Than-Human Design. *Design Issues* 36, 4 (2020), 33–44. doi:10.1162/desi_a_00612

[10] Hiroshi Ishii and Brygg Ullmer. 1997. Tangible Bits: Towards Seamless Interfaces between People, Bits and Atoms. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems (CHI '97)*. ACM, 234–241. doi:10.1145/258549.258715

[11] Brett Jones, Rajinder Sodhi, Michael Murdock, Ravish Mehra, Hrvoje Benko, Andrew Wilson, Eyal Ofek, Blair MacIntyre, Nikunj Raghuvanshi, and Lior Shapira. 2014. RoomAlive: Magical Experiences Enabled by Scalable, Adaptive Projector-Camera Units. In *Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology (UIST '14)*. ACM, 637–644. doi:10.1145/2642918.2647383

[12] Sergi Jordà, Günter Geiger, Marcos Alonso, and Martin Kaltenbrunner. 2007. The reacTable: Exploring the Synergy between Live Music Performance and Tabletop Tangible Interfaces. In *Proceedings of the 1st International Conference on Tangible and Embedded Interaction (TEI '07)*. ACM, 139–146. doi:10.1145/1226969.1226998

[13] Martin Kaltenbrunner and Ross Bencina. 2007. reacTIVision: A Computer-Vision Framework for Table-Based Tangible Interaction. In *Proceedings of the 1st International Conference on Tangible and Embedded Interaction (TEI '07)*. ACM, 69–74. doi:10.1145/1226969.1226983

[14] Benjamin Lee, Michael Sedlmair, and Dieter Schmalstieg. 2023. Design Patterns for Situated Visualization in Augmented Reality. *IEEE Transactions on Visualization and Computer Graphics* 30, 1 (2023). doi:10.1109/TVCG.2023.3327398

[15] Hanchuan Li, Can Ye, and Alanson P. Sample. 2015. IDSense: A Human Object Interaction Detection System Based on Passive UHF RFID. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15)*. ACM, 2555–2564. doi:10.1145/2702123.2702178

[16] Biswaksen Patnaik, Huaishu Peng, and Niklas Elmqvist. 2024. VisTorch: Interacting with Situated Visualizations Using Handheld Projectors. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. ACM, 1–13. doi:10.1145/3613904.3642857

[17] Krithik Ranjan, S. Sandra Bae, Peter Gyory, Ellen Yi-Luen Do, Clement Zheng, and Rong-Hao Liang. 2026. Why (Not) ReacTIVision: Emerging Challenges and Opportunities for Building Tangible User Interfaces with Computer Vision Toolkits. In *Proceedings of the Twentieth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '26)*. Association for Computing Machinery, New York, NY, USA, Article 30, 20 pages. doi:10.1145/3731459.3773326

[18] Kadek Ananta Satriadi, Andrew Cunningham, Ross T. Smith, Tim Dwyer, Adam Drogemuller, and Bruce H. Thomas. 2023. ProxSituated Visualization: An Extended Model of Situated Visualization Using Proxies for Physical Referents. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM. doi:10.1145/3544548.3580952

[19] Martin Schmitz, Mohammadreza Khalilbeigi, Matthias Balwierz, Roman Lissermann, Max Mühlhäuser, and Jürgen Steimle. 2015. Capricate: A Fabrication Pipeline to Design and 3D Print Capacitive Touch Sensors for Interactive Objects. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology (UIST '15)*. ACM, 253–258. doi:10.1145/2807442.2807503

[20] Martin Schmitz, Florian Müller, Max Mühlhäuser, Jan Riemann, and Huy Viet Viet Le. 2021. Itsy-Bits: Fabrication and Recognition of 3D-Printed Tangibles with Small Footprints on Capacitive Touchscreens. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) *(CHI '21)*. Association for Computing Machinery, New York, NY, USA, Article 419, 12 pages. doi:10.1145/3411764.3445502

[21] Sungbok Shin, Andrea Batch, Peter W. S. Butcher, Panagiotis D. Ritsos, and Niklas Elmqvist. 2024. The Reality of the Situation: A Survey of Situated Analytics. *IEEE Transactions on Visualization and Computer Graphics* 30, 8 (2024), 5147–5164. doi:10.1109/TVCG.2023.3285546

[22] Adalberto L. Simeone, Eduardo Velloso, and Hans Gellersen. 2015. Substitutional Reality: Using the Physical Environment to Design Virtual Reality Experiences. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15)*. ACM, 3307–3316.

doi:10.1145/2702123.2702389

[23] Katherine W. Song, Fiona Bell, Himani Deshpande, Ilan Mandel, Tiffany Wun, Mirela Alistar, Leah Buechley, Wendy Ju, Jeeeun Kim, Eric Paulos, Samar Sabie, and Ron Wakkary. 2024. Sustainable Unmaking: Designing for Biodegradation, Decay, and Disassembly. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. ACM, 1–7. doi:10.1145/3613905.3636300

[24] Martin Spindler, Christian Tominski, Heidrun Schumann, and Raimund Dachselt. 2010. Tangible Views for Information Visualization. In *ACM International Conference on Interactive Tabletops and Surfaces (ITS '10)*. ACM, 157–166. doi:10.1145/1936652.1936684

[25] Jeroen Spoelstra. 2022. What Is Life-Centered Design? https://www.unbeatenstudio.com/blog/what-is-life-centered-design. Blog post, Unbeaten Studio. Accessed July 28, 2026.

[26] Toni Steimle and Dieter Wallach. 2022. *Collaborative UX Design* (2 ed.). dpunkt.verlag. 240 pages. http://www.collaborative-uxdesign.com/

[27] Martin Tomitsch and Steve Baty. 2023. *Designing Tomorrow: Strategic Design Tactics to Change Your Practice, Organisation, and Planetary Impact.* BIS Publishers.

[28] Brygg Ullmer, Hiroshi Ishii, and Robert J. K. Jacob. 2005. Token+Constraint Systems for Tangible Interaction with Digital Information. *ACM Transactions on Computer-Human Interaction* 12, 1 (2005), 81–118. doi:10.1145/1057237.1057242

[29] Simon Voelker, Kosuke Nakajima, Christian Thoresen, Yuichi Itoh, Kjell Ivar Øvergård, and Jan Borchers. 2013. PUCs: Detecting Transparent, Passive Untouched Capacitive Widgets on Unmodified Multi-touch Displays. In *Proceedings of the 2013 ACM International Conference on Interactive Tabletops and Surfaces (ITS '13)*. ACM, 101–104. doi:10.1145/2512349.2512791

[30] Zeyu Yan, Su Hwan Hong, Josiah Hester, Tingyu Cheng, and Huaishu Peng. 2025. DissolvPCB: Fully Recyclable 3D-Printed Electronics Using Liquid Metal Conductors and PVA Substrates. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*. ACM, 1–17. doi:10.1145/3746059.3747604